Database Design and Develop Project

Professor Navarro

COP3710 – Database 1

Background

An information system performs three sets of services:

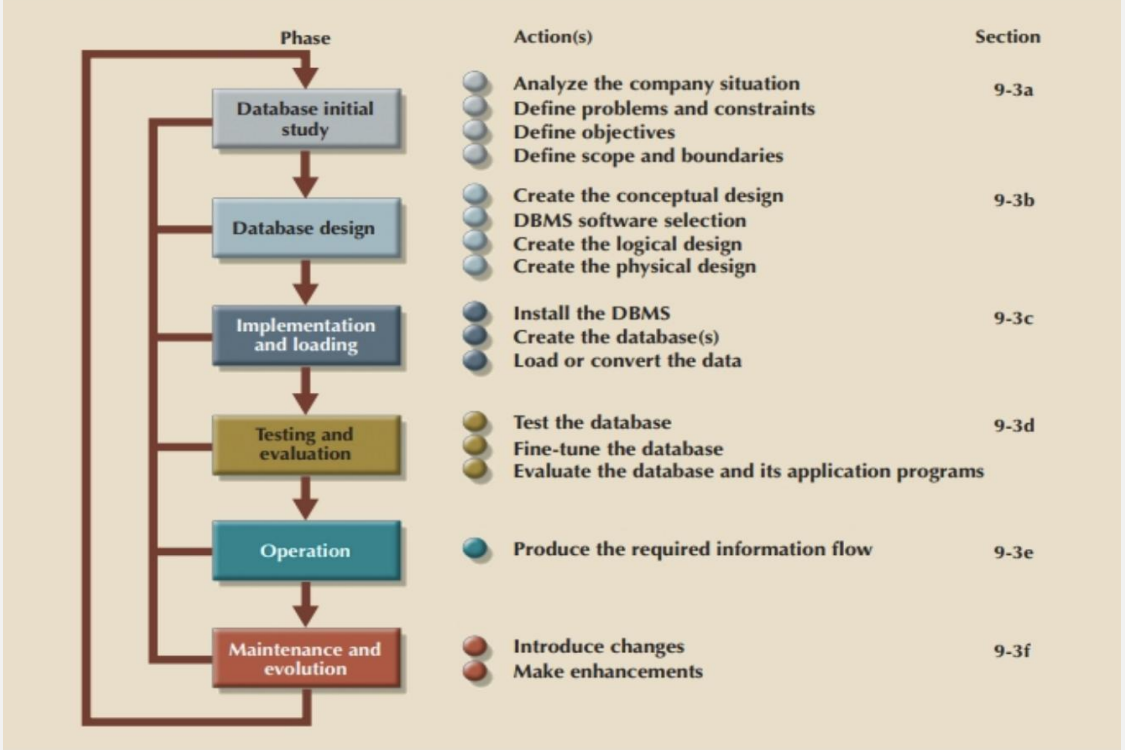
- It provides for data collection, storage, and retrieval.
- It facilitates the transformation of data into information.
- It provides the tools and conditions to manage both data and information

A database is a fact (data) repository that serves an information system. If the database is designed poorly, the data/information transformation may be less than successful, while is it reasonable to expect trouble with efficient and capable management of data and information. Good database design lays the foundation for a successful information system. Database design must yield a database that:

- Does not fall prey to uncontrolled data duplication, thus preventing data anomalies and the attendant lack of data integrity.
- Is efficient in its provision of data access.
- Serves the needs of the information system.

Due to the relationship between the two, database life cycle development can have the same phases as the information system software development life cycle. You will follow the DBLC in this project.

FIGURE 9.3 THE DATABASE LIFE CYCLE (DBLC)



In this project, you can work solo or on a team to analyze the requirements for, design, implement, document, and demonstrate a relational database system for a proposal to fictional retail company. You should select specific products (physical or virtual) for your project to sale and describe specific application requirements. You will implement a database where your system must maintain a current record of inventory that is updated to reflect the transactions performed. You must keep track of customers, including their personal information, and purchase history.

Procedure

Plan a fictitious retail company, determine the products they sale, what problems they may have and propose a database in order to solve their business problem.

1. Define the problem(s).
2. Create the conceptual design.
3. Verify the conceptual design with all user views.
4. Define all system processes and data constraints.

5. Assure that the database design will comply with all achievable end user requirements.

Decide on Top-down (start with entities) or Bottom-up Design (start by defining the required attributes)

Create a proposal (word document) with **subheadings** on/in each page:

- Cover page with Business name, team members, user name, DB name
- Pg1- Executive summary of business, problem, and solution (1 paragraph)
- Pg2- Define Problems and Constraints, Define objectives in more detail (half to full page)
- Pg3- Define scope and boundaries with business rules (half to full page)
- Pg4- EERD with Crow's Foot format (includes keys, relationships, attributes)
    - Entities should include the sales channel which will require 12 minimum entities for:
        - Products (at least 3 subtypes), Inventory, Invoice, Customer (include personal information that could be shared with Customer Relationship Management software), Product Vendor at the minimum
        - Add additional entities to show invoice (time variant) history
        - When determining invoice total, use derived values. Views are acceptable.
        - Identify keys and necessary attributes (on average 5 minimum attributes each)
        - At least one auto-generated surrogate key should be used
        - Use auto-increment or sequences and not-null at least once
    - Indicate relationship types, cardinality, completeness, disjoint or overlap
    - At least five (5) entities should have a minimum of 500 rows
    - Use http://filldb.info or similar for extra dummy sample data
    - Each table on average should have minimum of 20 real data rows
    - Data model should be normalized to 3NF unless a justified business reason.
- Pg5- SQL dump of (show only first page in report) database (mysqldump)
- Pg6- Output of a sql query to show a specific invoice based on a customer name and date of purchase, also include SQL code
- Pg7- Output of sql query to show a current inventory of all products sold within a date range, also include SQL code
- Pg8- Output of sql query to show a current inventory of all products that match one specific product type, also include SQL code
- Pg9- Output of sql stored procedure that allows user to pass in a value and in order to join three tables to produce an output (you decide the attributes), also include SQL code
- Pg10- SQL code used to insert new product into product table (show all attributes)
- Pg11- SQL code to generate a new invoice, you decide product and customer information
- Pg12- Output of a virtual table (of your choice), also include SQL code
- Pg13 – Output of sql query that demonstrates a frequency distribution of customers who purchased items that cost more than x order by price of purchased item in descending order limit to only 3 results. Demonstrate an aggregate function in your output also (ie min, max, avg)
- Pg14 – SQL code to demonstrate a sub query with arithmetic operations and aliases.

- Pg15 – SQL code that joins five (5) tables and returns rows that match multiple conditions using Boolean. Display any column you decide. Analyze the time to run the query, then add an index (or composite index). Re-analyze the time for the query to determine if improvements were made. Show your results
- Pg16- Show data dictionary of tables (see fig 1)
- Pg17 -Summary of your database solution as conclusion to business proposal (minimum 2-3 paragraphs)
- Pg18 – Individual paragraph from each team member on what you learned personally.
- Pg19 - Table that shows the percentage of effort by each team member and responsibilities

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1. Create ERD
2. Create Database (MariaDB)
3. Add Data
4. Test the database and fine tune
5. Make improvements as necessary

Submit SQL dump of entire database (mysqldump) as a single file with extension ".sql".

Quality of written presentation, power point document and physical presentation will be evaluated. This is a final project and should be detailed and professional. The database design should be efficient and normalized properly to 3rd normal form. The database dump will be accessed by your professor.

Summary of Canvas submission documents

1. Proposal Word Document with 18 page minimum
2. Presentation video will be submitted only to the Project Presentation Assignment in Canvas.
3. SQL dump file of database with extension ".sql"
4. No Zip files

Fig 1.

MariaDB>
```
SELECT table_name, column_name, column_type, is_nullable,
column_comment FROM information_schema.COLUMNS WHERE table_schema =
'cnavarro_db_aircotest' ORDER BY table_name, ordinal_position ASC;
```

Fig 2. Data Dictionary Example

```
MariaDB [cnavarro_db_aircotest]> SELECT     table_name,     column_name,     col
umn_type,     is_nullable,     column_comment  FROM     information_schema.COLUM
NS  WHERE     table_schema = 'cnavarro_db_aircotest'  ORDER BY     table_name,
   ordinal_position ASC;
+-------------+---------------+-------------+-------------+----------------+
| table_name  | column_name   | column_type | is_nullable | column_comment |
+-------------+---------------+-------------+-------------+----------------+
| EMPLOYEE    | EMP_NUM       | int(11)     | YES         |                |
| EMPLOYEE    | EMP_LNAME     | varchar(15) | YES         |                |
| EMPLOYEE    | EMP_FNAME     | varchar(50) | YES         |                |
| EMPLOYEE    | EMP_INITIAL   | varchar(50) | YES         |                |
| EMPLOYEE    | EMP_HIRE_DATE | datetime    | YES         |                |
| EMPLOYEE    | EMP_TYPE      | varchar(50) | YES         |                |
| EMPLOYEE_V1 | EMP_NUM       | int(11)     | YES         |                |
| EMPLOYEE_V1 | EMP_LNAME     | varchar(15) | YES         |                |
```
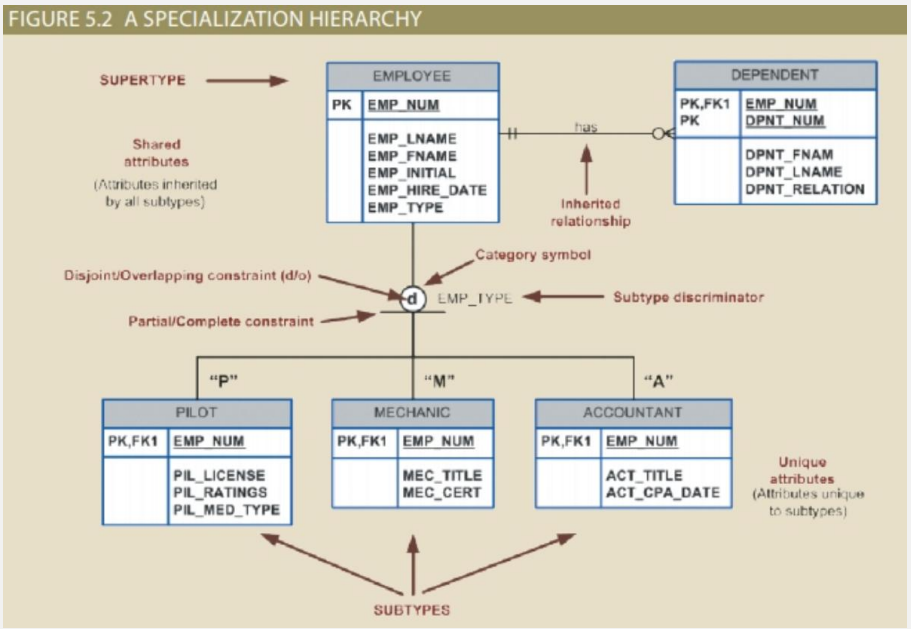
Fig 5.2 ERD Example